Typical real-world examples used to explain stacks usually include a pile of dirty dishes (you pick from the top of the pile and so you first wash the ones that have been added more recently). It is in computer science, however, that stacks find their largest and more crucial uses:

- Memory management allowing your programs to run (the call stack).
- Implementation of the memento pattern, in turn allowing to undo actions in your editor or browsing back and forth your browser's history.
- Enabling recursion and being used as support for several algorithms.
- In the JavaScript world, you might have heard of time-travel in React.

And the list above is by no means to be considered exhaustive.

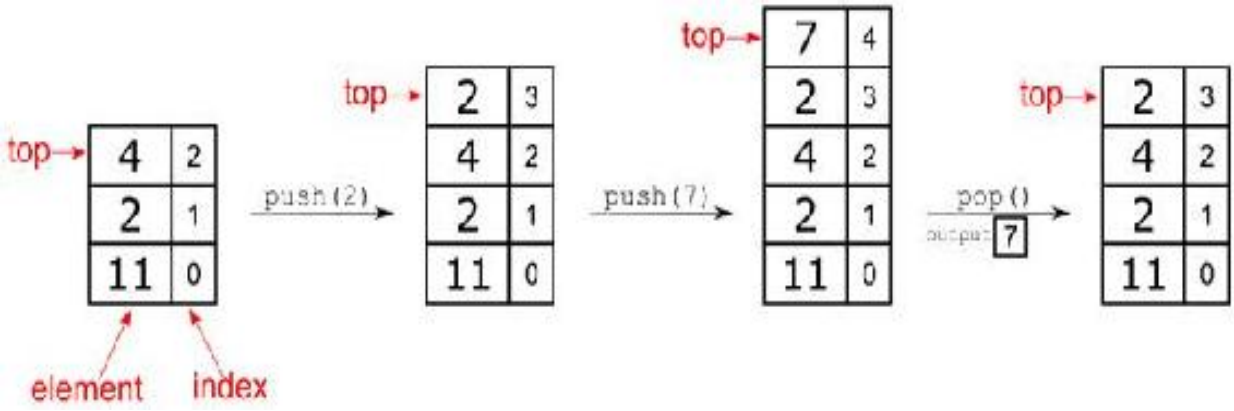


Figure D.2 Operations on a stack. Insertion (`push`) adds element to top of the stack. Delete (`pop`) remove the top element and returns it. For stacks (as well as for queues) a `peek` operation is also usually available,